CDNs

DATA SHEET

COUPLING / DECOUPLING NETWORKS

FOR IMMUNITY TESTING ACCORDING TO IEC / EN 61000-4-6

Frankonia EMC Test-Systems GmbH

Daimlerstr. 17

91301 Forchheim

Germany

+49 9191 73666-0

+49 9191 73666-20

sales@frankonia-emv.com

WWW.FRANKONIA-SOLUTIONS.COM

FRANKONIA
EMC Test-Systems GmbH

IEC / EN 61000-4-6
IEC 61326-3-2, Ne-21

- **Coupling networks for the following frequencies:**
  150 kHz – 230 MHz:
  Standard IEC / EN 61000-4-6

- **Coupling networks for the following frequencies:**
  10 kHz – 230 MHz:
  Standard IEC / EN 61000-4-6
  IEC 61326-3-2, NE-21

OVERVIEW

Coupling/Decoupling Networks (CDN) for immunity testing against conducted disturbances, induced by radio-frequency fields according to IEC / EN 61000-4-6 and other standards.

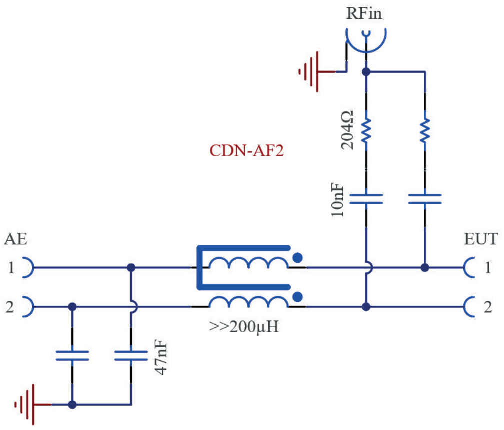
A range of different types is available for various applications.
Custom models can also be manufactured upon customer request.

TYPES	APPLICATION AREA	EXAMPLES
CDN-M	Unshielded power supply lines	AC mains, DC in industrial installations, grounding connection
CDN-AF	Unshielded, unbalanced interconnection lines	Audio cables PLC lines
CDN-T CDN-RJ	Unshielded, balanced interconnection lines	Communication lines
CDN-CAN	BUS lines	CAN
CDN-S CDN-RJ45-S CDN-USB CDN-HDMI CDN-Firewire	Shielded interconnection lines	HDMI, coaxial cables, cables for LAN and USB connections

FRANKONIA
EMC Test-Systems GmbH

CDN-AF: Coupling / decoupling of disturbance signals onto unshielded, unbalanced lines**

MODEL DESIGNATION	CDN-AF2/AF3/ AF4/AF5/AF6/AF8	CDN-AF2-10K/ AF3-10K/ AF5-10K
RF In		
Frequency range (RF In)	150 kHz – 230 MHz	10 kHz – 230 MHz
Power (RF In)	6 W (100 % ED)	6 W (100 % ED)
Decoupling attenuation (RF In – AE)	20 dB (150 kHz – 230 MHz) 40 dB (1 MHz – 100 MHz)	20 dB (150 kHz – 230 MHz) 40 dB (1 MHz – 100 MHz)
Insertion loss (RF In – EUT)	10 dB ± 1 dB (150 kHz – 80 MHz) 10 dB + 3 dB (150 kHz – 230 MHz)	10 dB ± 1 dB (150 kHz – 80 MHz) 10 dB + 3 dB (150 kHz – 230 MHz)
Connection	N	N
EUT / AE		
Operating voltage	100 VAC / 150 VDC	100 VAC / 150 VDC
Rated current (AE – EUT)	5 A	5 A
Through attenuation (AE – EUT)	< 1 dB (DC – 100 kHz)	< 1 dB (DC – 100 kHz)
Connection	2 mm safety sockets	2 mm safety sockets
General data		
Dimensions (W × H × D)	160 x 82 x 240 mm	160 x 82 x 240 mm



CIRCUIT DIAGRAM CDN-AF2

Daimlerstr. 17, 91301 Forchheim +49 9191 73666-0 +49 9191 73666-20 sales@frankonia-emv.com

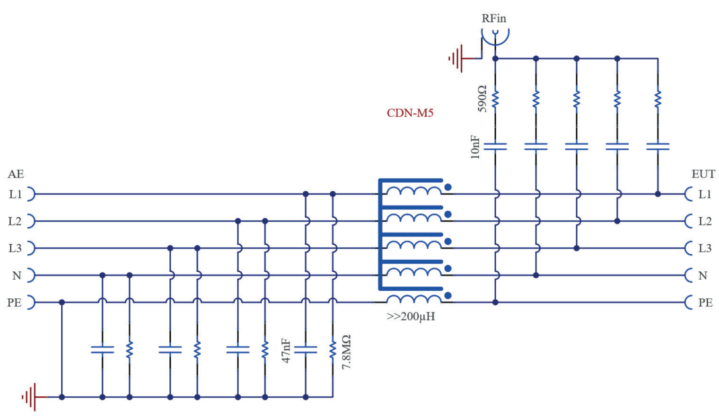
FRANKONIA
EMC Test-Systems GmbH

CDN-M:
- Coupling / decoupling of disturbance signals for unshielded power supply lines
- Rated voltage up to 1000 V
- Rated current up to 125 A

GENERAL CHARACTERISTICS	10 kHz-Version	150 kHz-Version	125 A-Version
Power (RF In)	6 W (100 % ED)	6 W (100 % ED)	6 W (100 % ED)
Decoupling attenuation (RF In – AE)	30 dB (150 kHz – 80 MHz) 20 dB (80 MHz – 230 MHz)	30 dB (150 kHz – 80 MHz) 15 dB (80 MHz – 230 MHz)	30 dB (150 kHz – 80 MHz)
Insertion loss (RF In – EUT)	10 dB +2/–1 dB (150 kHz – 80 MHz) 10 dB + 5 dB (80 MHz – 230 MHz)	10 dB +2/–1 dB (150 kHz – 80 MHz) 10 dB + 5 dB (80 MHz – 230 MHz)	10 dB +2/–1 dB (150 kHz – 80 MHz)
Coaxial connector	Coaxial N (female)	Coaxial N (female)	Coaxial N (female)
Through attenuation (AE – EUT)	< 1 dB (DC – 100 kHz)	< 1 dB (DC – 100 kHz)	< 1 dB (DC – 100 kHz)
Connection (AE / EUT)	4 mm MC safety sockets	4 mm MC safety sockets	Push-in spring terminal, max. 35 mm²
Dimensions (W × H × D)	160 × 102 × 240 mm	160 × 102 × 240 mm	200 × 122 × 400 mm
	Rated current	**Rated voltage**	**Pin assignment**
CDN M-Types 10 kHz - 230 MHz			
CDN M1-10k	32A (max. 0,5A Isat.)	500V AC / 1000V DC	PE
CDN M2-10k	32A	500V AC / 1000V DC	L, N
CDN M2+M3-10k	32A	500V AC / 1000V DC	L, N, PE
CDN M3-10k	32A	500V AC / 1000V DC	L, N, PE
CDN M3-L-10k	32A	500V AC / 1000V DC	L1, L2, L3
CDN M4-10k	32A	500V AC / 1000V DC	L1, L2, L3, N
CDN M5-10k	32A	500V AC / 1000V DC	L1, L2, L3, N, PE

FRANKONIA
EMC Test-Systems GmbH

	Rated current	Rated voltage	Pin assignment
CDN M-Types 150 kHz – 230 MHz			
CDN M1	32A	500V AC / 1000V DC	PE
CDN L1	16A	500V AC / 1000V DC	L
CDN L1-32A	32A	500V AC / 1000V DC	L
CDN M2	32A	500V AC / 1000V DC	L, N
CDN M2+M3	32A	500V AC / 1000V DC	L, N, PE
CDN M3-L	32A	500V AC / 1000V DC	L1, L2, L3
CDN M4	32A	500V AC / 1000V DC	L1, L2, L3, PE
CDN M4-N	32A	500V AC / 1000V DC	L1, L2, L3, N
CDN M5	32A	500V AC / 1000V DC	L1, L2, L3, N, PE
CDN M-Types 125A, 150 kHz – 80 MHz			
CDN M2-125A	125A	1000V AC / 1000V DC	L, N
CDN M3-125A	125A	1000V AC / 1000V DC	L, N, PE
CDN M3-L-125A	125A	1000V AC / 1000V DC	L1, L2, L3
CDN M4-125A	125A	1000V AC / 1000V DC	L1, L2, L3, PE
CDN M4-N-125A	125A	1000V AC / 1000V DC	L1, L2, L3, N
CDN M5-125A	125A	1000V AC / 1000V DC	L1, L2, L3, N, PE



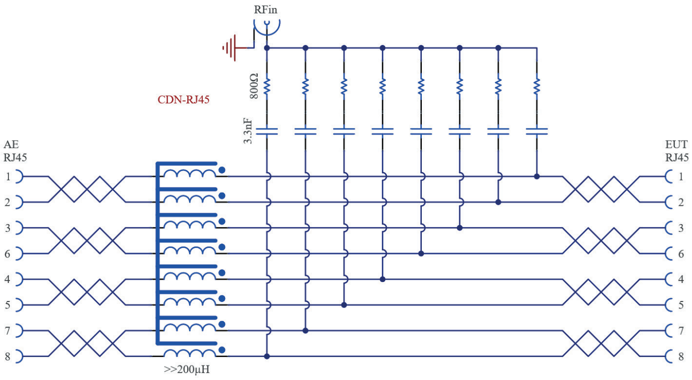
CIRCUIT DIAGRAM CDN-M5

Daimlerstr. 17, 91301 Forchheim +49 9191 73666-0 +49 9191 73666-20 sales@frankonia-emv.com

FRANKONIA
EMC Test-Systems GmbH

CDN-RJ: Coupling / decoupling of disturbance signals onto unshielded, balanced lines

MODEL DESIGNATION	CDN-RJ45	CDN-RJ45-10K
RF In		
Frequency range (RF In)	150 kHz – 230 MHz	10 kHz – 230 MHz
Power (RF In)	6 W (continuous power)	6 W (continuous power)
Decoupling attenuation (RF In – AE)	20 dB (150 kHz – 230 MHz)	20 dB (150 kHz – 230 MHz)
Insertion loss (RF In – EUT)	10 dB ± 1 dB (150 kHz – 80 MHz) 10 dB + 3 dB (80 MHz – 230 MHz)	10 dB ± 1 dB (150 kHz – 80 MHz) 10 dB + 3 dB (80 MHz – 230 MHz)
Connection	N	N
EUT / AE	100 VAC / 150 VDC	100 VAC / 150 VDC
Operating voltage	1,5 A	1,5 A
Rated current (AE – EUT)	< 1 dB (DC – 10 kHz)	< 1 dB (DC – 10 kHz)
Through attenuation (AE – EUT)	< 10 dB (10 MHz – 100 MHz)	< 10 dB (10 MHz – 100 MHz)
Data rate	Up to 1 Gbit/s	Up to 1 Gbit/s
Connector	RJ45 socket (8-pin)	RJ45 socket (8-pin)
General data		
Dimensions (W × H × D)	160 x 82 x 240 mm	160 x 82 x 240 mm



CIRCUIT DIAGRAM CDN-RJ45

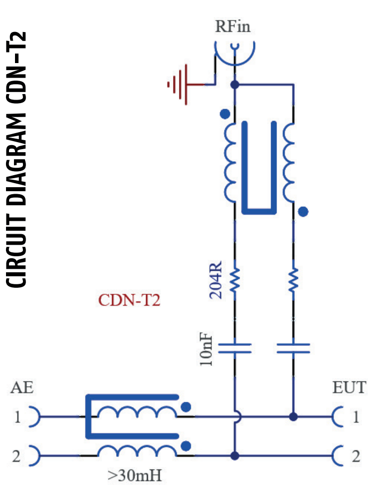
Daimlerstr. 17, 91301 Forchheim +49 9191 73666-0 +49 9191 73666-20 sales@frankonia-emv.com

FRANKONIA
EMC Test-Systems GmbH

CDN-T: Coupling / decoupling of disturbance signals onto unshielded, balanced lines

MODEL DESIGNATION	CDN-T2, -T4, -T8	CDN-T2-10K, -T4-10K, -T8-10K
RF In		
Frequency range (RF In)	150 kHz – 230 MHz	10 kHz – 230 MHz
Power (RF In)	6 W (continuous power)	6 W (continuous power)
Decoupling attenuation (RF In – AE)	20 dB (150 kHz – 230 MHz)	20 dB (150 kHz – 230 MHz)
Insertion loss (RF In – EUT)	10 dB ± 1 dB (150 kHz – 230 MHz)	10 dB ± 1 dB (150 kHz – 230 MHz)
Connection	N	N
EUT / AE		
Operating voltage AC	100 V	100 V
Operating voltage DC	150 V	150 V
Rated current (AE – EUT)	0,5 A	0,5 A
Through attenuation (AE – EUT)	< 1 dB (DC – 1 MHz) < 10 dB (1 MHz – 100 MHz)	< 1 dB (DC – 1 MHz) < 10 dB (1 MHz – 100 MHz)
Connection	2 mm safety socket	2 mm safety socket
General data		
Dimensions (W × H × D)	160 x 82 x 240 mm	160 x 82 x 240 mm

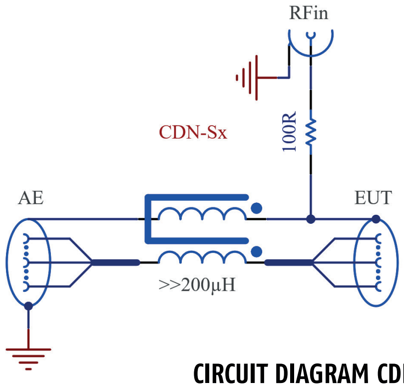
CIRCUIT DIAGRAM CDN-T2



Daimlerstr. 17, 91301 Forchheim +49 9191 73666-0 +49 9191 73666-20 sales@frankonia-emv.com

CDN-S: Coupling / decoupling of disturbance signals onto shielded lines

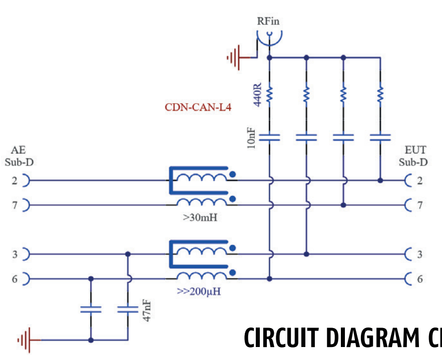
MODEL DESIGNATION	CDN-S1	-S1-75	-S2	-S3	-S4	-S9	-S15	-S25
EUT / AE								
Connection EUT	BNC Socket	BNC Socket	XLR Plug	XLR Plug	5-pin XLR Plug	9-pin Sub-D Plug	15-pin Sub-D Plug	25-pin Sub-D Plug
Connection AE	BNC Socket	BNC Socket	XLR Socket	XLR Socket	5-pin XLR Socket	9-pin Sub-D Socket	15-pin Sub-D Socket	25-pin Sub-D Socket
Operating voltage	150 VAC / 200 VDC							
Rated current (AE – EUT)	1,5 A							
Through attenuation (AE – EUT)	< 1 dB (0 – 10 MHz) < 10 dB (10 MHz – 500 MHz)							
RF In								
Frequency range (RF In)	10 kHz – 230 MHz							
Power (RF In)	6 W (continuous power)							
Decoupling attenuation (RF In – AE)	>35 dB (150 kHz – 80 MHz) >30 dB (80 MHz – 230 MHz)							
Insertion loss (RF In – EUT)	10 dB ± 1 dB (150 kHz – 80 MHz); 10 dB + 3 dB (80 MHz – 230 MHz)							
Connection	N							
General data								
Dimensions (W × H × D)	160 mm x 82 mm x 240 mm							



CIRCUIT DIAGRAM CDN-SX

CDN-CAN: Coupling / decoupling of disturbance signals for unshielded, balanced lines

MODEL DESIGNATION	CDN-CAN-L4	CDN-CAN-L5
RF In		
Frequency range (RF In)	150 kHz – 230 MHz	150 kHz – 230 MHz
Power (RF In)	6 W (continuous power)	6 W (continuous power)
Decoupling attenuation (RF In – AE)	PIN 2+7: > 35 dB (150 kHz – 230 MHz) PIN 3+9: > 35 dB (150 kHz – 200 MHz) >25 dB (200 MHz – 230 MHz)	PIN 2+7: > 35 dB (150 kHz – 230 MHz) PIN 3+6+9: > 35 dB (150 kHz – 200 MHz) >25 dB (200 MHz – 230 MHz)
Insertion loss (RF In – EUT)	10 dB ± 1 dB (150 kHz – 230 MHz)	10 dB ± 1 dB (150 kHz – 230 MHz)
Connection	N	N
EUT / AE		
Operating voltage AC	50 V	50 V
Operating voltage DC	50 V	50 V
Rated current (AE – EUT)	PIN 2+7 = 0,5 A; PIN 3+9 = 3 A	PIN 2+7 = 0,5 A PIN 3+6+9 = 3 A
Through attenuation (AE – EUT)	PIN 2+7: < 1 dB (DC – 10 MHz) < 10 dB (10 MHz – 500 MHz) PIN 3+9: < 1 dB (DC – 100 kHz)	PIN 2+7: < 1 dB (DC – 10 MHz), < 10 dB (10 MHz – 500 MHz) PIN 3+6+9: < 1 dB (DC – 100 kHz)
Connection	9-pin Sub-D Socket	9-pin Sub-D Socket
General data		
Dimensions (W × H × D)	160 x 82 x 240 mm	160 x 82 x 240 mm



CIRCUIT DIAGRAM CDN-CAN-L4

Daimlerstr. 17, 91301 Forchheim +49 9191 73666-0 +49 9191 73666-20 sales@frankonia-emv.com

FRANKONIA
EMC Test-Systems GmbH

CDN-RJ45-S / -HDMI 2.1 / -FIREWIRE: Coupling / decoupling of disturbance signals onto shielded lines

MODEL DESIGNATION	CDN-RJ45-S	CDN-HDMI (HDMI 2.1)	CDN-FIREWIRE
RF In			
Frequency range (RF In)	10 kHz – 230 MHz	10 kHz – 230 MHz	10 kHz – 230 MHz
Power (RF In)	6 W (continuous power)	6 W (continuous power)	6 W (continuous power)
Decoupling attenuation (RF In – AE)	>30 dB (150 kHz – 230 MHz)	50 dB (150 kHz – 80 MHz) 25 dB (80 MHz – 230 MHz)	
Insertion loss (RF In – EUT)	10 dB ± 1 dB (150 kHz – 80 MHz) 10 dB + 3 dB (80 MHz – 230 MHz)		
Connection	N	N	N
EUT / AE			
Operating voltage AC	100 VAC / 150 VDC	100 VAC / 150 VDC	100 VAC / 150 VDC
Rated current (AE – EUT)	1,0 A	0,5 A	0,5 A
Through attenuation (AE – EUT)	< 0,3 dB (DC – 10 MHz) < 1 dB (10 MHz – 100 MHz)	< 1 dB (DC – 10 MHz) < 10 dB (10 MHz – 500 MHz)	
Data rate	< 3 dB (100 MHz – 500 MHz)		
Connector	Up to 10 Gbit/s	Shielded HDMI sockets	FireWire socket, 6-pin
General data	Shielded RJ45 socket, 8-pin		
Dimensions (W × H × D)	160 x 82 mm x 240 mm		

FRANKONIA
EMC Test-Systems GmbH

CDN-USB- 3.0 / -Z / -P: Coupling / decoupling of disturbance signals onto shielded lines (USB 2.0 compatible)
CDN-USB TYP-C: Coupling / decoupling of disturbance signals onto shielded lines

MODEL DESIGNATION	CDN-USB-3.0-Z	CDN-USB-3.0-P	CDN-USB TYPE-C
RF In			
Frequency range (RF In)	10 kHz – 230 MHz	10 kHz – 230 MHz	10 kHz – 230 MHz
Power (RF In) continuous power	6 W	6 W	6 W
Decoupling attenuation (RF In – AE)	>50 dB (150 kHz – 80 MHz) >25 dB (80 MHz – 230 MHz)		>50 dB (150 kHz – 80 MHz) >25 dB (80 MHz – 230 MHz)
Insertion loss (RF In – EUT)	10 dB ± 1 dB (150 kHz – 80 MHz) 10 dB + 3 dB (80 MHz – 230 MHz)		10 dB ± 1 dB (150 kHz – 80 MHz) 10 dB + 3 dB (80 MHz – 230 MHz)
Connection	N	N	N
EUT / AE			
Operating voltage AC	100 VAC / 150 VDC	100 VAC / 150 VDC	100 VAC / 150 VDC
Rated current (AE – EUT)	0,9 A	0,9 A	0,9 A
Through attenuation (AE – EUT)	< 1 dB (DC – 10 MHz) < 10 dB (10 MHz – 500 MHz)		< 1 dB (DC – 10 MHz) < 10 dB (10 MHz – 500 MHz)
Connector	EUT: USB-B AE: USB-A	UT: USB-A AE: USB-B	EUT: USB-C Socket AE: USB-C Socket
General data			
Dimensions (W × H × D)	160 x 82 x 240 mm	160 x 82 x 240 mm	160 x 82 x 240 mm

Daimlerstr. 17, 91301 Forchheim +49 9191 73666-0 +49 9191 73666-20 sales@frankonia-emv.com

Frankonia EMC Test-Systems GmbH
Daimlerstr. 17
91301 Forchheim
Germany

+49 9191 73666-0
+49 9191 73666-20
sales@frankonia-emv.com